1 Extremes, Analytic functions, Series

1.1 Extremes

Theorem 1.1: Extremes of holomorphic functions

Ω connected and bounded, f holomorphic on Ω. Then $|f|$ does not have local extremes unless it is constant.

For nondegenerate extremes. $f(z) = u(x,y) + iv(x,y)$. $|f|^2 = f \cdot \overline{f} = u^2 + v^2 = g$. $Dg = 0$ and $D^2 g$ can't be negatively-definite.

$$g_x = \frac{\partial}{\partial x}(u^2 + v^2) = 2uu_x + 2vv_x$$

$$g_y = \frac{\partial}{\partial y}(u^2 + v^2) = 2uu_y + 2vv_y.$$

If $(g_x, g_y) = (0,0)$ then $(u,v) = 0$ or from Cauchy Riemann

$$\det \begin{bmatrix} u_x & v_x \\ u_y & v_y \end{bmatrix} = 0 \implies u_x = v_x = u_y = v_y = 0 \implies f'(x) = 0$$

$$\operatorname{tr} D^2 g = g_{xx} + g_{yy} = 2u(u_{xx} + u_{uu}) + 2v(v_{xx} + v_{yy}) = 0,$$

which is a contradiction. □

Proof. Take a circle γ around the point maximizing $|f|$:

$$f(z_0) = \frac{1}{2\pi i}\int_\gamma \frac{f(w)}{w - z_0} dw = \frac{1}{2\pi}\int_0^{2\pi} f(z_0 + re^{it})$$

This is the average of f along γ, and we get

$$|f(z_0)| \leqslant \max_{z \in \gamma} |f(z)|.$$

But from maximality, $|f|$ must be constantly equal $|f(z_0)|$ on γ. Analogously, it is constant inside any circle (contained in Ω) around z_0. By some standard topology i is constant everywhere as Ω is open and connected.

□

1.2 Analytic functions and Series

Theorem 1.2: Holomorphic $\implies$ Analytic

Any holomorphic function is analytic.

Proof. Take a circle γ around z_0, $w \in \gamma$. For z in the interior of γ (to make the series converge), we have

$$\frac{1}{w - z} = \frac{1}{w - z_0} \sum_{n=0}^{\infty} \left(\frac{z - z_0}{w - z_0}\right)^n,$$

hence

$$f(z) = \frac{1}{2\pi i}\int_\gamma f(w)\sum_{n=0}^{\infty}\frac{(z-z_0)^n}{(w-z_0)^{n+1}}dw =$$
$$\sum \frac{f^{(n)}(z_0)}{n!}(z-z_0)^n.$$

We can exchange integration with summation, since the inside is uniformly convergent on a compact set. □

Corollary: if f is analytic around 0, then

$$f(w) = \sum a_n(z-z_0)^n$$

where $|a_n| < \frac{M}{R^n}$ for an upper bound M of $|f|$ inside the considered disk. Note that R can be taken to be maximal possible. Consider arctan. Its convergence radius at 0 is 1, so it can't extend to an entire function. In this case, it has poles at $\pm i$.

Theorem 1.3: Analytic $\implies$ Holomorphic

Any analytic function is holomorphic.

Proof. This is a standard bound-bash. The same argument as for real analytic functions. □

Lemma 1.4: Hadamard lemma

$f\colon \mathbb{R}\to\mathbb{R}$. $f(0)=0$. If $f\in C^k$, then $\exists g\in C^{k-1}$ such that $f(x)=xg(x)$.

Proof.

$$f(x) = \int_0^1 \frac{d}{dt}f(xt)dt = \int_0^1 xf'(xt)dt = x\int_0^1 f'(xt)dt$$

□

Theorem 1.5: degree of vanishing

If f is holomorphic, then it can be uniquely written as $z^n g(z)$ where $n\geqslant 0$ and $g(z)\neq 0$.

Proof. Look at the first nonzero term in the power series expansion. □

Corollary: if f is holomorphic and $f(0)=0$, then $\frac{f(z)}{z}$ is also holomorphic.